

Ratio, proportion and percentages - Set 1

TMUA practice questions by topic. Specification M3.1-M3.11. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The price of a coat is reduced by 20% in a sale, and then the sale price is reduced by a further 25%. The final price is what percentage of the original price?
- A 50%
 - B 55%
 - C 65%
 - D 60%
 - E 40%
 - F 45%
- 2 After a 15% increase, a salary is £27 600. What was the original salary?
- A £23 000
 - B £31 740
 - C £23 460
 - D £24 500
 - E £24 000
 - F £25 000
- 3 £600 is divided between A , B and C in the ratio $2 : 3 : 7$. How much more does C receive than A ?
- A £300
 - B £250
 - C £150
 - D £200
 - E £350
 - F £100

- 4 y is inversely proportional to the square of x . When $x = 2$, $y = 9$. Find y when $x = 6$.
- A** 3
B 4
C 1
D 81
E 27
F $\frac{1}{3}$
- 5 A bacteria population doubles every 3 hours. By what factor does it grow in 24 hours?
- A** 16
B 48
C 64
D 128
E 8
F 256
- 6 Two similar solids have surface areas in the ratio 4 : 9. Find the ratio of their volumes.
- A** 12 : 27
B 16 : 81
C 2 : 3
D 8 : 27
E 64 : 729
F 4 : 9
- 7 A mixture contains sand and cement in the ratio 5 : 2. How much cement must be added to 35 kg of the mixture to make the ratio 5 : 3?
- A** 2.5 kg
B 15 kg
C 3 kg
D 7 kg
E 5 kg
F 10 kg

- 8** £2000 is invested at 5% compound interest per year. Which expression gives the amount after n years?
- A** 2000×5^n
 - B** $2000 + 100n$
 - C** 2000×1.05^n
 - D** 2000×1.5^n
 - E** $2000(1 + 0.05n)$
 - F** 2000×0.05^n
- 9** The value of a car falls by 20% each year. After how many complete years is it first worth less than half its original value?
- A** 5
 - B** 6
 - C** 4
 - D** 10
 - E** 2
 - F** 3
- 10** A recipe for 12 biscuits uses 150 g of flour and 100 g of butter. Ann has 1 kg of flour and 500 g of butter. What is the greatest number of biscuits she can make?
- A** 80
 - B** 60
 - C** 50
 - D** 48
 - E** 66
 - F** 72

11 Which of the following statements is/are true?

I Increasing a quantity by 50% and then decreasing the result by 50% returns it to its original value.

II A 10% increase followed by a 10% increase is a 21% increase overall.

III If A is 25% more than B , then B is 20% less than A .

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

12 A quantity increases by $x\%$ and then decreases by $x\%$. The overall change is a decrease of 4%. Find x .

A 2

B 20

C 10

D 40

E 4

F 25